

# Relation Algebra

## §14. Thinning Algorithms

### §14.1. Thinning

(14.1a)

For  $Q : A \rightarrow A$ ,  $\text{thin}(Q) \triangleq (\exists \exists) \cap (\epsilon \setminus (Q^\circ \epsilon)) : EA \rightarrow EA$  (8.1).

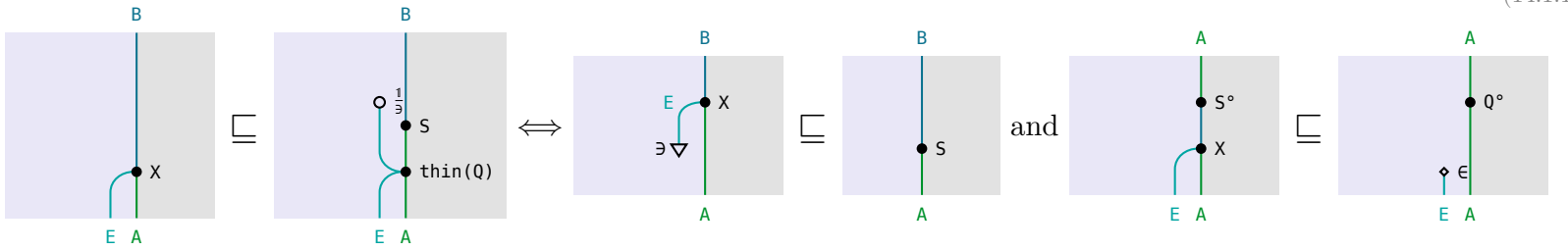
$ys \text{ thin}(Q) xs \iff xs \subseteq ys \wedge (\forall a \in ys. \exists b \in xs. b \ Q \ a)$  the same  $^\circ$  as  $\triangleq \text{est}$ :  $\epsilon X$  runs  $a \rightarrow ys \rightarrow xs$ , member of  $ys$  first, and  $Q^\circ \epsilon$  runs  $a \rightarrow b \rightarrow xs$ , so  $(a, b) \in Q^\circ$  reads  $b \ Q \ a$ ; at  $Q \triangleq \leq$  every  $a \in ys$  keeps some  $b \leq a$  in  $xs$ , the end  $\text{est}(\leq)$  picks

(14.1b)

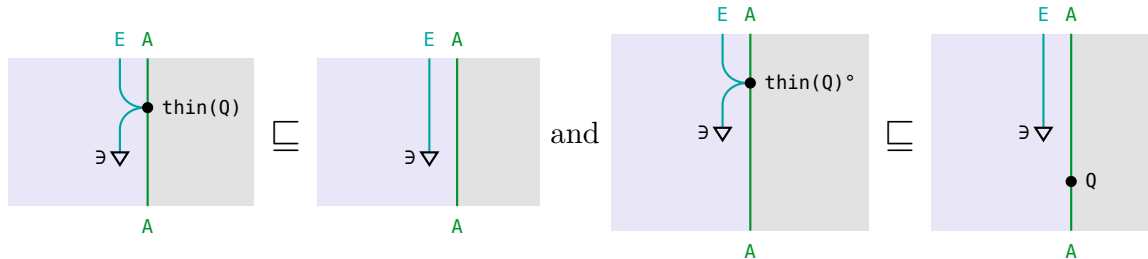
| the law                                                                                                                                                                                                                                                                        | what it says                                                                                                          |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| $X \subseteq \frac{S}{\exists} \text{thin}(Q) \iff X \exists \subseteq S$ and $S^\circ X \subseteq Q^\circ \epsilon$                                                                                                                                                           | everything kept is an S-value, and every S-value has a Q-lower bound among the kept ones                              |
| $\text{thin}(Q) \exists \subseteq \exists$                                                                                                                                                                                                                                     | everything a thinning keeps was in the set                                                                            |
| $\text{thin}(Q)^\circ \exists \subseteq \exists Q$                                                                                                                                                                                                                             | every element of the set has a Q-lower bound among the kept ones                                                      |
| $Q \subseteq R \implies \text{thin}(Q) \subseteq \text{thin}(R)$                                                                                                                                                                                                               | the fewer pairs Q relates, the fewer subsets count as thinnings                                                       |
| $1 \subseteq Q \implies 1 \subseteq \text{thin}(Q)$<br>$Q \subseteq Q \implies \text{thin}(Q) \text{thin}(Q) \subseteq \text{thin}(Q)$                                                                                                                                         | keeping everything is always a legal thinning                                                                         |
| $Q \subseteq R \wedge 1 \subseteq Q \wedge R R \subseteq R \implies \text{est}(R) = \text{thin}(Q) \text{est}(R)$ $Q \subseteq R, 1 \subseteq Q, R R \subseteq R$ — (14.1.2a); weaker than the book's "both preorders": Q transitive is never used and $1 \subseteq R$ follows | <b>thin-introduction</b> : thinning first cannot lose an R-minimum                                                    |
| $\text{thin}(Q) \supseteq \text{est}(Q) \frac{1}{\exists}$ (8.2) — (14.1.2d)                                                                                                                                                                                                   | <b>thin-elimination</b> : keeping one element is a thinning, but its domain is the sets $\text{est}(Q)$ is defined on |
| $\frac{S}{\exists} \text{thin}(Q) \supseteq \frac{S}{\exists} \text{est}(R) \frac{1}{\exists}$<br>(8.3), $R \cap (S^\circ S) \subseteq Q$ — thin-elimination variant                                                                                                           | the usable variant: R need only refine Q between values S gives one argument                                          |
| $P(\text{thin}(Q)) \text{union} \subseteq \text{union thin}(Q)$ (8.4)                                                                                                                                                                                                          | thinning each member set is a thinning of the union                                                                   |

#### §14.1.1. $X \subseteq (S^\circ \exists) \text{thin}(Q) \iff X \exists \subseteq S$ and $S^\circ X \subseteq Q^\circ \epsilon$

(14.1.1a)



(14.1.1b)



(14.1.1a) at  $X \triangleq \text{thin}(Q)$ ,  $S \triangleq \exists$ : the left side is  $\text{thin}(Q) \subseteq \text{thin}(Q)$ , since  $\exists \exists = \wedge(\exists) = 1$

#### §14.1.2. $\text{est}(R) = \text{thin}(Q) \text{est}(R)$ given $Q \subseteq R$ , $1 \subseteq Q$ , $R R \subseteq R$

(14.1.2a)

**$\text{est}(R) \sqsubseteq \text{thin}(Q) \text{ est}(R)$**   
 the  $\sqsubseteq$  half: keeping everything is a thinning —  $1 \sqsubseteq Q$

$EA \text{—} \boxed{\text{est}(R)} \text{—} A$   
 $\sqsubseteq$   
 $EA \text{—} \boxed{\text{thin}(Q)} \text{—} \boxed{\text{est}(R)} \text{—} A$   
 $\sqsubseteq$   
 $1 \sqsubseteq \text{thin}(Q) \text{ — (14.1b), } Q \text{ reflexive}$

(14.1.2b)

**$\text{thin}(Q) \text{ est}(R) \sqsubseteq \exists$**   
 the  $\sqsubseteq$  half, first condition of the UP of  $\text{est}$  at  $X \triangle \text{thin}(Q) \text{ est}(R)$  — est-up

$EA \text{—} \boxed{\text{thin}(Q)} \text{—} \boxed{\text{est}(R)} \text{—} A$   
 $\sqsubseteq$   
 $EA \text{—} \boxed{\text{thin}(Q)} \text{—} \boxed{\exists} \text{—} A$   
 $\sqsubseteq$   
 $\text{est}(R) \sqsubseteq \exists \text{ — est-laws first row at } X \triangle \text{est}(R)$   
 $\sqsubseteq$   
 $EA \text{—} \boxed{\exists} \text{—} A$   
 $\sqsubseteq$   
 $\text{thin}(Q) \exists \sqsubseteq \exists \text{ — (14.1.1a) at } S \triangle \exists, X \triangle \text{thin}(Q): \exists \exists = 1, \text{ so}$   
 $\text{the left side is } \text{thin}(Q) \sqsubseteq \text{thin}(Q)$

(14.1.2c)

**$\epsilon \text{ thin}(Q) \text{ est}(R) \sqsubseteq R^\circ$**   
 the  $\sqsubseteq$  half, second condition —  $Q \sqsubseteq R$ ,  $R$  transitive

$A \text{—} \boxed{\epsilon} \text{—} \boxed{\text{thin}(Q)} \text{—} EA \text{—} \boxed{\text{est}(R)} \text{—} A$   
 $\sqsubseteq$   
 $A \text{—} \boxed{Q} \text{—} \boxed{\epsilon} \text{—} EA \text{—} \boxed{\text{est}(R)} \text{—} A$   
 $\sqsubseteq$   
 $A \text{—} \boxed{Q} \text{—} \boxed{R} \text{—} A$   
 $\sqsubseteq$   
 $A \text{—} \boxed{R} \text{—} \boxed{R} \text{—} A$   
 $\sqsubseteq$   
 $A \text{—} \boxed{R} \text{—} A$

$\epsilon \text{ thin}(Q) \sqsubseteq Q^\circ \epsilon \text{ — (14.1.1a) at } S \triangle \exists, X \triangle \text{thin}(Q), \text{ second half}$   
 $\epsilon \text{ est}(R) \sqsubseteq R^\circ \text{ — est-laws first row at } X \triangle \text{est}(R), \text{ second half, conversed}$   
 $Q \sqsubseteq R, \text{ conversed}$   
 $R \text{ transitive, conversed}$

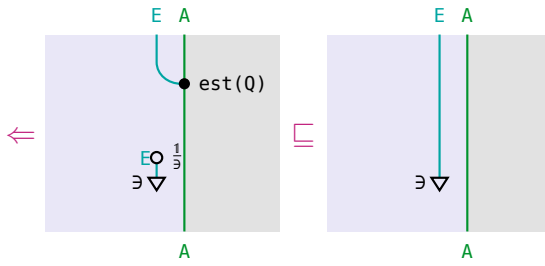
$$\text{est}(Q) \stackrel{1}{\exists} \text{thin}(Q)$$

(14.1.2d)

 the singleton holding a  $Q$ -least member of a set is a thinning of that set

$$\text{est}(Q) \stackrel{1}{\exists} \exists \exists$$

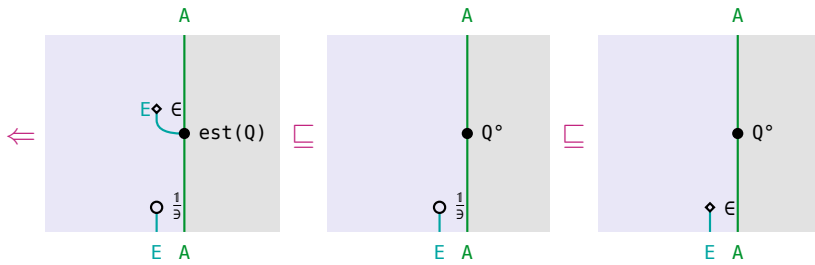
every member of the singleton is a member of the set



$$\begin{aligned} \Leftarrow & EA \text{---} \boxed{\text{est}(Q)} \text{---} A \text{---} \boxed{\frac{1}{\exists}} \text{---} EA \text{---} \boxed{\exists} \text{---} A \\ \sqsubseteq & EA \text{---} \boxed{\exists} \text{---} A \end{aligned}$$

 the  $\exists/\exists$  half of  $\trianglelefteq$  thin at  $X \trianglelefteq \text{est}(Q) \stackrel{1}{\exists}$ 
 $\frac{1}{\exists} \exists = 1$  — pow-laws — then  $\text{est}(Q) \sqsubseteq \exists$  — est-laws

$$\epsilon \text{est}(Q) \stackrel{1}{\exists} \sqsubseteq Q^\circ \epsilon$$

 the singleton's member stands in  $Q$  to every member of the set


$$\begin{aligned} \Leftarrow & A \text{---} \boxed{\epsilon} \text{---} EA \text{---} \boxed{\text{est}(Q)} \text{---} A \text{---} \boxed{\frac{1}{\exists}} \text{---} EA \\ \sqsubseteq & A \text{---} \boxed{Q} \text{---} \boxed{\frac{1}{\exists}} \text{---} EA \\ \sqsubseteq & A \text{---} \boxed{Q} \text{---} \boxed{\epsilon} \text{---} EA \end{aligned}$$

 the  $\epsilon \backslash (Q^\circ \epsilon)$  half of  $\trianglelefteq$  thin,  $\neg \sqsubseteq Q^\circ \epsilon$ 
 $\epsilon \text{est}(Q) \sqsubseteq Q^\circ$  — est-up at  $X \trianglelefteq \text{est}(Q)$ , conversed

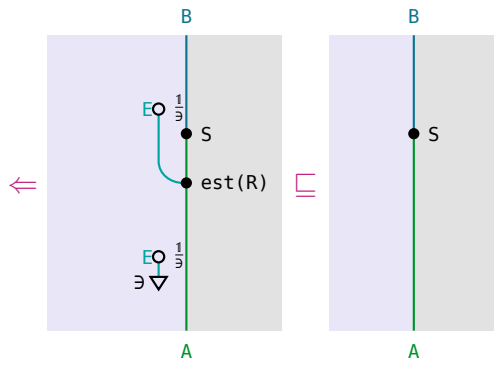
 $\frac{1}{\exists} \sqsubseteq \epsilon$ , since  $\frac{1}{\exists} \exists = 1$  with  $\frac{1}{\exists}$  a map — pow-laws

$$\text{Rn}S^\circ \sqsubseteq Q \Rightarrow \frac{S}{S} \text{est}(R) \frac{1}{S} \sqsubseteq \frac{S}{S} \text{thin}(Q)$$

(14.1.2e)

 given  $\text{Rn}(S^\circ S) \sqsubseteq Q$ ,  $Q$  a preorder

$$\frac{S}{S} \text{est}(R) \frac{1}{S} \ni S$$

 every member of the singleton is a value  $S$  returns


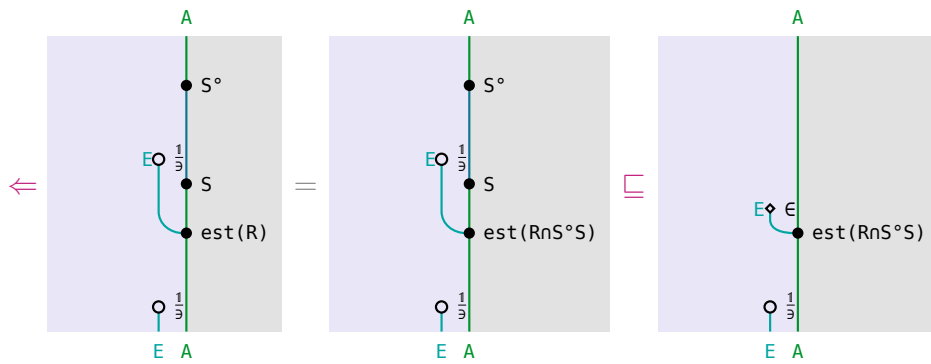
$$\leftarrow B \xrightarrow{\frac{1}{S}} EB \xrightarrow{E(S)} EA \xrightarrow{\text{est}(R)} A \xrightarrow{\frac{1}{S}} EA \xrightarrow{\ni} A$$

 the first of (14.1b)' two conditions at  $X \triangleq \frac{S}{S} \text{est}(R) \frac{1}{S}$ 

$$\sqsubseteq B \xrightarrow{\frac{1}{S}} A$$

 $\frac{1}{S} \ni 1$  — pow-laws — then  $\frac{S}{S} \text{est}(R) = \text{Sn}(S^\circ \setminus R^\circ) \sqsubseteq S$  — est-laws

$$\text{Rn}S^\circ \sqsubseteq Q \Rightarrow S^\circ \frac{S}{S} \text{est}(R) \frac{1}{S} \sqsubseteq Q^\circ \in$$

 the singleton's member stands in  $Q$  to every value  $S$  returns at the same argument


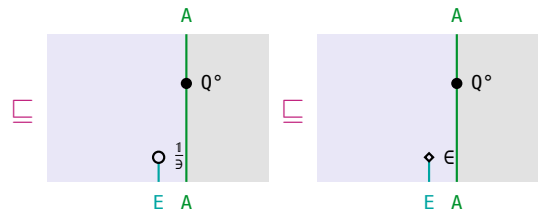
$$\leftarrow A \xrightarrow{\frac{1}{S}} B \xrightarrow{\frac{1}{S}} EB \xrightarrow{E(S)} EA \xrightarrow{\text{est}(R)} A \xrightarrow{\frac{1}{S}} EA$$

 the second condition,  $\neg Q^\circ \in$ 

$$= A \xrightarrow{\frac{1}{S}} B \xrightarrow{\frac{1}{S}} EB \xrightarrow{E(S)} EA \xrightarrow{\text{est}(RnS^\circ S)} A \xrightarrow{\frac{1}{S}} EA$$

 $\frac{S}{S} \text{est}(R) = \frac{S}{S} \text{est}(RnS^\circ S)$  — est-laws

$$\sqsubseteq A \xrightarrow{\frac{1}{S}} EA \xrightarrow{\text{est}(RnS^\circ S)} A \xrightarrow{\frac{1}{S}} EA$$

 $S^\circ \frac{S}{S} \sqsubseteq \in$ , since  $\frac{S}{S} \ni S$  with  $\frac{S}{S}$  a map — pow-laws


$$\sqsubseteq A \xrightarrow{Q} \frac{1}{S} \xrightarrow{\frac{1}{S}} EA$$

 $\in \text{est}(RnS^\circ S) \sqsubseteq (RnS^\circ S)^\circ$  — est-up at  $X \triangleq \text{est}(RnS^\circ S)$ , conversed — then  $\text{Rn}(S^\circ S) \sqsubseteq Q$ 

$$\sqsubseteq A \xrightarrow{Q} \frac{1}{S} \xrightarrow{\frac{1}{S}} EA$$

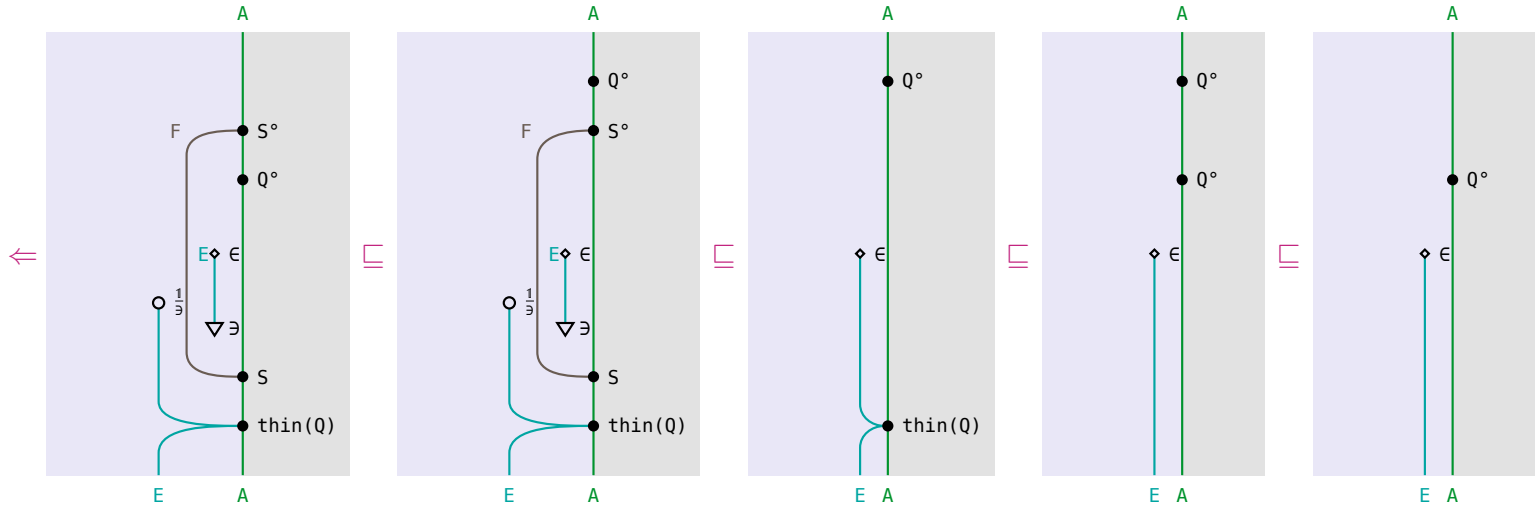
 $\frac{1}{S} \sqsubseteq \in$ , since  $\frac{1}{S} \ni 1$  with  $\frac{1}{S}$  a map — pow-laws

$$\text{PreservesRecip}(F) \wedge Q \subseteq Q \wedge \text{MonoAlg}(S, Q) \Rightarrow \langle \frac{F(\exists)S}{\exists} \text{thin}(Q) \rangle \sqsubseteq \langle \frac{(S)}{\exists} \text{thin}(Q) \rangle \quad (14.1.2f)$$

thinning at every step of the reduce is a thinning of the whole candidate set —  $S$  monotonic on  $Q$ ,  $Q$  a preorder

$$\langle \frac{F(\exists)S}{\exists} \text{thin}(Q) \rangle \sqsubseteq \langle \frac{(S)}{\exists} \text{thin}(Q) \rangle \quad \text{and} \quad \langle \frac{(S)}{\exists} \rangle \circ \langle \frac{F(\exists)S}{\exists} \text{thin}(Q) \rangle \sqsubseteq Q^\circ \epsilon$$

(14.1b) at  $X \triangleq \langle \frac{F(\exists)S}{\exists} \text{thin}(Q) \rangle$ ,  $\langle \frac{(S)}{\exists} \rangle$  for its  $S$



$$\leftarrow A \rightarrow \boxed{S} \rightarrow FA \rightarrow \boxed{F(Q^\circ \epsilon)} \rightarrow FEA \rightarrow \boxed{\frac{1}{\exists}} \rightarrow EFEA \rightarrow \boxed{E(F(\exists)S)} \rightarrow \boxed{\text{thin}(Q)} \rightarrow EA$$

the first by fusion; hylomorphism theorem at the bound  $Q^\circ \epsilon$  reduces the second to  $\neg \epsilon Q^\circ \epsilon$

$$\sqsubseteq A \rightarrow \boxed{Q} \rightarrow \boxed{S} \rightarrow FA \rightarrow \boxed{F(\epsilon)} \rightarrow FEA \rightarrow \boxed{\frac{1}{\exists}} \rightarrow EFEA \rightarrow \boxed{E(F(\exists)S)} \rightarrow \boxed{\text{thin}(Q)} \rightarrow EA$$

$S^\circ F(Q^\circ) \sqsubseteq Q^\circ S^\circ$  — mon-str at  $S$ , conversed;  $F(R)^\circ = F(R^\circ)$  — relator-laws

$S^\circ F(\epsilon) \xrightarrow{\frac{F(\exists)S}{\exists}} \epsilon$ , since  $\frac{F(\exists)S}{\exists} \sqsubseteq F(\exists)S$  with  $\frac{F(\exists)S}{\exists}$  a map — pow-laws

$\epsilon \text{thin}(Q) \sqsubseteq Q^\circ \epsilon$ , the  $\epsilon \setminus (Q^\circ \epsilon)$  half of  $\triangle \text{thin}$  — adj-all

$Q^\circ Q^\circ = Q^\circ$ ,  $Q$  a preorder

$$\sqsubseteq A \rightarrow \boxed{Q} \rightarrow \boxed{\epsilon} \rightarrow \boxed{\text{thin}(Q)} \rightarrow EA$$

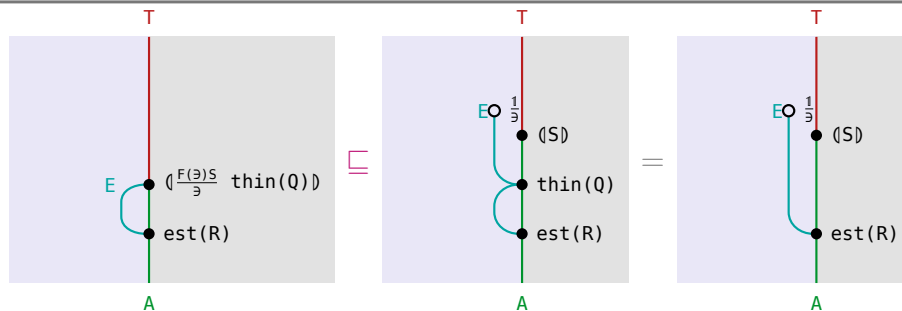
$$\sqsubseteq A \rightarrow \boxed{Q} \rightarrow \boxed{Q} \rightarrow \boxed{\epsilon} \rightarrow EA$$

$$\sqsubseteq A \rightarrow \boxed{Q} \rightarrow \boxed{\epsilon} \rightarrow EA$$

$$\text{PreservesRecip}(F) \wedge Q \subseteq R \wedge 1 \sqsubseteq Q \wedge Q \subseteq Q \wedge R \subseteq R \wedge \text{MonoAlg}(S, Q) \Rightarrow \langle \frac{F(\exists)S}{\exists} \text{thin}(Q) \rangle \text{est}(R) \sqsubseteq \langle \frac{(S)}{\exists} \text{thin}(Q) \rangle \quad (14.1.2g)$$

$\text{est}(R)$

the thinning fold refines the optimisation problem itself —  $S$  monotonic on  $Q$ ,  $Q \subseteq R$ , both preorders



$$T \rightarrow \boxed{FEA} \rightarrow \boxed{\frac{1}{\exists}} \rightarrow EFEA \rightarrow \boxed{E(F(\exists)S)} \rightarrow \boxed{\text{thin}(Q)} \rightarrow EA \rightarrow \boxed{\text{est}(R)} \rightarrow A$$

$$\sqsubseteq T \rightarrow \boxed{\frac{1}{\exists}} \rightarrow E(T) \rightarrow \boxed{E(\langle \frac{(S)}{\exists} \rangle)} \rightarrow \boxed{\text{thin}(Q)} \rightarrow EA \rightarrow \boxed{\text{est}(R)} \rightarrow A$$

thinning theorem

$$= T \rightarrow \boxed{\frac{1}{\exists}} \rightarrow E(T) \rightarrow \boxed{E(\langle \frac{(S)}{\exists} \rangle)} \rightarrow EA \rightarrow \boxed{\text{est}(R)} \rightarrow A$$

$$\text{est}(R) = \text{thin}(Q) \text{ est}(R) \quad (14.1b), Q \subseteq R$$

## §14.2. Paths in a layered network

(14.2a)

$F(A, X) = A + A \times X$ ,  $L = \text{list}^+$  with initial algebra  $\alpha \triangleq [\text{wrap}, \text{cons}] : F(A, LA) \rightarrow LA$ .

$\text{wrapz} \triangleq \langle \text{wrap}, \text{zero} \rangle$ ,  $\text{consw}(a, (xs, n)) = (\text{cons}(a, xs), \text{wt}(a, \text{head}(xs)) + n)$ .

$\text{cost} \triangleq \langle [\text{wrapz}, \text{consw}] \rangle \pi_2$ ,  $\langle [\text{wrapz}, \text{consw}] \rangle = \langle 1, \text{cost} \rangle$ ,  $R \triangleq \text{cost} \leq \text{cost}^\circ$ .

$Q \triangleq R \cap (\text{head} \text{ head}^\circ)$ ,  $S \triangleq F(1, \exists) \alpha$ ,  $\frac{F(\exists, 1)}{\exists} = 1 + \text{cpl}$ ,  $\frac{F(1, \exists)}{\exists} = 1 + \text{cpr}$ ,  $\text{step} \triangleq \text{cpr } P(\text{cons}) \text{ est}(R)$ .

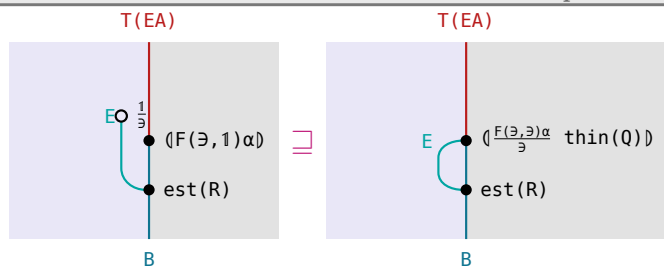
(14.2b)

| the law                                                                         | what it says                                                                                                                                                        |
|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $F(\exists, Q) \alpha \sqsubseteq F(\exists, 1) \alpha Q$                       | $F(\exists, 1) \alpha$ is monotonic on $Q$ ; on $R$ it is not, since the next edge can cost arbitrarily much                                                        |
| $S \text{ head} \sqsubseteq [1, \pi_1] \quad S \triangleq F(1, \exists) \alpha$ | $S \text{ head}$ is simple, which gives $Rn(S \circ S) \sqsubseteq Q$ : between two paths $S$ builds from one argument, equal cost and equal head already means $Q$ |

$$\text{PreservesRecip}(F) \wedge Q \sqsubseteq R \wedge 1 \sqsubseteq Q \wedge Q \sqsubseteq Q \wedge R \circ R \circ \sqsubseteq R \circ \wedge \text{MonoAlg}(F(\exists, 1) \alpha, Q) \wedge Rn(F(1, \exists) \alpha) \circ F(1, \exists) \alpha \sqsubseteq Q \quad (14.2c)$$

$$\Rightarrow \langle \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists) \alpha}{\exists} \text{est}(R)) \rangle \text{est}(R) \sqsubseteq \langle \frac{F(\exists, 1) \alpha}{\exists} \rangle \text{est}(R)$$

a least-cost path in a layered network, as a fold over the layers



$$\begin{aligned} & T(EA) \rightarrow \boxed{\frac{1}{\exists}} \rightarrow E(T(EA)) \rightarrow E(F(\exists, 1) \alpha) \rightarrow EB \rightarrow \text{est}(R) \rightarrow B \\ & \sqsubseteq T(EA) \rightarrow \boxed{\text{FEB}} \rightarrow \boxed{\frac{1}{\exists}} \rightarrow EFEB \rightarrow E(F(\exists, \exists) \alpha) \rightarrow \text{thin}(Q) \rightarrow EB \rightarrow \text{est}(R) \rightarrow B \end{aligned}$$

 (14.1.2g), at  $F(\exists, 1) \alpha$  monotonic on  $Q$  — (14.2b).

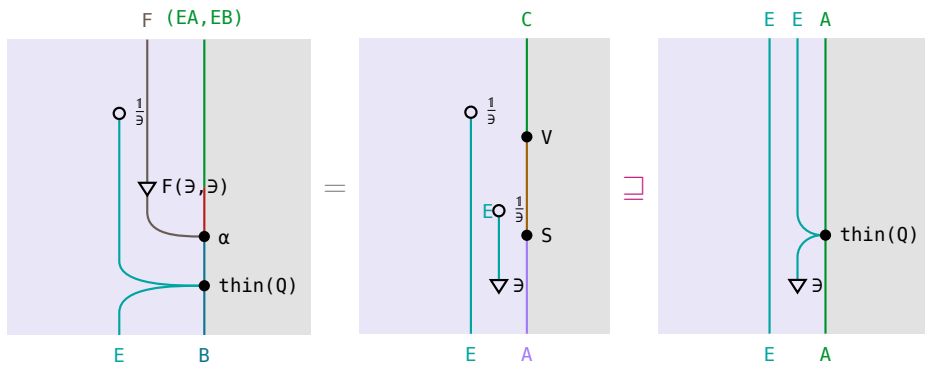
 $\sqsubseteq$  (14.2d) under  $\langle \rangle$  monotonic: the whole chain runs inside the reduce, and the  $\text{est}(R)$  behind it never moves.

$$Rn(F(1, \exists)\alpha) \circ F(1, \exists)\alpha \sqsubseteq Q \Rightarrow \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)\alpha}{\exists} \text{est}(R)) \sqsubseteq \frac{F(\exists, \exists)\alpha}{\exists} \text{thin}(Q) \quad (14.2d)$$

thinning the algebra of a layered network costs no more than taking the program's two cases

$$\frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)\alpha}{\exists}) P(\text{thin}(Q)) \text{union} \sqsubseteq \frac{F(\exists, \exists)\alpha}{\exists} \text{thin}(Q)$$

thinning each set the algebra's parts build, then taking the union, is below thinning the one set the whole algebra builds



$$\begin{aligned} & F(EA, EB) \xrightarrow{\frac{1}{\exists}} E(F(EA, EB)) \xrightarrow{E(F(\exists, \exists)\alpha)} \text{thin}(Q) \xrightarrow{} EB \\ & = C \xrightarrow{\frac{1}{\exists}} EC \xrightarrow{E(V)} EW \xrightarrow{P(\frac{S}{\exists})} EEA \xrightarrow{\text{union}} EA \\ & \sqsubseteq EEA \xrightarrow{P(\text{thin}(Q))} \text{union} \xrightarrow{} EA \end{aligned}$$

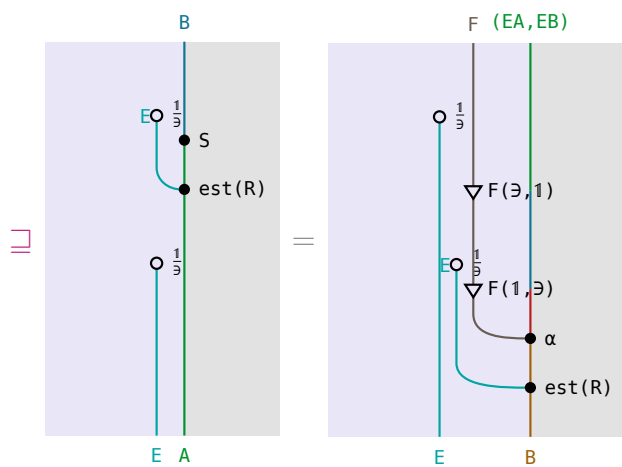
the algebra of (14.2c) row 2

$$F(\exists, \exists) = F(\exists, 1)F(1, \exists); \quad \frac{F(\exists, 1)F(1, \exists)\alpha}{\exists} = \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)\alpha}{\exists}) \text{union}.$$

$$\text{union thin}(Q) \sqsupseteq P(\text{thin}(Q)) \text{union} \quad (14.1b).$$

$$Rn(F(1, \exists)\alpha) \circ F(1, \exists)\alpha \sqsubseteq Q \Rightarrow \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)\alpha}{\exists} \text{est}(R)) \sqsubseteq \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)\alpha}{\exists}) P(\text{thin}(Q)) \text{union}$$

choosing an R-least member of each of those sets is below thinning each and taking the union



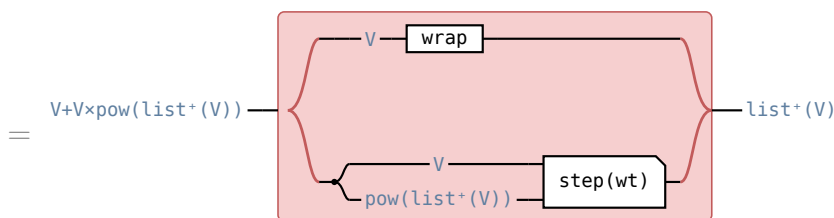
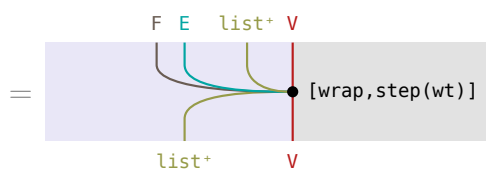
$$\begin{aligned} & B \xrightarrow{\frac{1}{\exists}} EB \xrightarrow{E(S)} EA \xrightarrow{\text{est}(R)} A \xrightarrow{\frac{1}{\exists}} EA \\ & = F(EA, EB) \xrightarrow{\frac{1}{\exists}} E(F(EA, EB)) \xrightarrow{E(F(\exists, 1))} E(FA, EB) \xrightarrow{P(\frac{F(1, \exists)\alpha}{\exists} \text{est}(R))} EB \end{aligned}$$

$$\frac{S}{\exists} \text{thin}(Q) \sqsupseteq \frac{S}{\exists} \text{est}(R) \xrightarrow{\frac{1}{\exists}} \quad (14.1b) \text{ at } S \triangleq F(1, \exists)\alpha, Rn(S \circ S) \sqsubseteq Q \quad (14.2b).$$

$$P(\frac{1}{\exists}) \text{union} = 1.$$

$$cp(F, \text{list}^+(V)) P(\alpha) \text{est}(\text{path}(\text{wt})) = [\text{wrap}, \text{step}(\text{wt})]$$

at the layered network, choosing an R-least path from each case is the program's two cases  $[\text{wrap}, \text{step}]$



$$\alpha \text{ a map, so } \frac{F(1, \exists)\alpha}{\exists} = \frac{F(1, \exists)}{\exists} P(\alpha); \quad \frac{F(1, \exists)}{\exists} P(\alpha) \text{est}(R) = [\text{wrap}, \text{step}] \triangleq \text{cost}.$$

$$= \frac{F(\exists, 1)}{\exists} = 1 + \text{cpl} \triangleq \text{cost}$$

## §14.3. Implementing thin

(14.3a)

$\text{setify} : [A] \rightarrow EA$ ,  $\text{cup} : EA \times EA \rightarrow EA$ ,  $\text{cp}(F) \triangleq \frac{F(\exists)}{\exists}$ ,  $\text{listcp} : F(L) \rightarrow LF$ ,  $\text{sort}(P) \triangleq \text{setify}^\circ \text{ ordered } P$  for  $P$  a connected preorder.

$\text{thinlist}(Q)$  is any  $\text{thinlist}(Q) \sqsubseteq \text{subseq}$  with  $\text{thinlist}(Q) \text{ setify} \sqsubseteq \text{setify thin}(Q)$ ; one is  $\langle [\text{nil}, \text{bump } Q] \rangle$ ,  $\text{bump } Q (a, []) = [a]$ ,  $\text{bump } Q (a, [b] \# xs) = (b \text{ } Q \text{ } a \rightarrow [a] \# xs, a \text{ } Q \text{ } b \rightarrow [b] \# xs, [a] \# [b] \# xs)$ .

**Binary thinning** data:  $S = (f_1 p_1) \cup (f_2 p_2)$  with  $p_1, p_2$  coreflexive;  $Q$  a preorder with  $Q \sqsubseteq R$  and both  $f_1 p_1, f_2 p_2$  monotonic on  $Q$ ;  $P$  a connected preorder with both  $f_1, f_2$  monotonic on  $P$ ;  $g_i \triangleq \text{list}(f_i) \text{ filter}(p_i)$ .

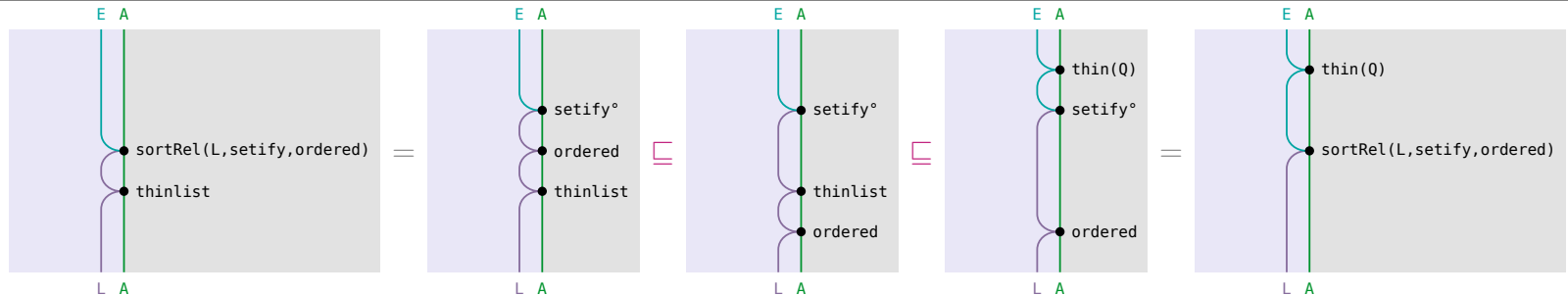
(14.3b)

| the law                                                                                                  | what it says                                                                                                       |
|----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| $\text{thinlist}(Q) \text{ xs} = [\text{minlist}(Q) \text{ xs}]$<br>(8.5), $Q$ connected, $xs$ non-empty | what thinning should come to when it can: one element                                                              |
| $\text{sort}(P) \text{ thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{ sort}(P)$ (8.6) — (14.3c)          | thinning a sorted list is a thinning of the set — this is what $\text{thinlist}(Q) \sqsubseteq \text{subseq}$ buys |
| $\text{sort}(P) \text{ minlist}(Q) \sqsubseteq \text{est}(Q)$ (8.7)                                      | a minimum of the sorted list is a minimum of the set                                                               |
| $\text{sort}(f P f^\circ) \text{ list}(f) \sqsubseteq P(f) \text{ sort}(P)$ (8.8)                        | shunt a function through a sort                                                                                    |
| $\text{sort}(P) \text{ filter}(p) \sqsubseteq E(p) \text{ sort}(P)$ (8.9), $p$ coreflexive               | filtering a sorted list sorts the restricted set                                                                   |
| $(\text{sort}(P) \times \text{sort}(P)) \text{ merge}(P) \sqsubseteq \text{cup sort}(P)$ (8.10)          | merging two sorted lists sorts their union                                                                         |
| $F(\text{sort}(P)) \text{ listcp} \sqsubseteq \text{cp}(F) \text{ sort}(FP)$<br>(8.11), $F$ linear       | $\text{listcp}$ is the list implementation of the cartesian product $\text{cp}(F)$                                 |

$\text{Map}(\text{setify}) \wedge \text{Coreflexive}(\text{ordered}) \wedge \text{thinlist} \sqsubseteq \text{subseq} \wedge \text{ordered subseq} \sqsubseteq \text{subseq ordered} \wedge$  (14.3c)

$\text{thinlist setify} \sqsubseteq \text{setify thin}(Q) \Rightarrow \text{sortRel}(L, \text{setify}, \text{ordered}) \text{ thinlist} \sqsubseteq \text{thin}(Q) \text{ sortRel}(L, \text{setify}, \text{ordered})$

a thinning of the sorted list lists a thinning of the set —  $P$  a connected preorder,  $\text{thinlist}(Q) \sqsubseteq \text{subseq}$ .



$EA \xrightarrow{\text{setify}} \text{ordered} \xrightarrow{\text{thinlist}} LA$

$= EA \xrightarrow{\text{setify}} \text{ordered} \xrightarrow{\text{thinlist}} LA$

$EA \xrightarrow{\text{setify}} \text{thinlist} \xrightarrow{\text{ordered}} LA$

$\sqsubseteq \text{ordered } P \text{ thinlist}(Q) \sqsubseteq \text{thinlist}(Q) \text{ ordered } P$ , since  $\text{thinlist}(Q) \sqsubseteq \text{subseq} \triangleq \text{thinlist}$  — and a subsequence of a  $P$ -ordered list is  $P$ -ordered

$EA \xrightarrow{\text{thin}(Q)} \text{setify} \xrightarrow{\text{ordered}} LA$

$\triangleq \text{thinlist's thinlist}(Q) \text{ setify} \sqsubseteq \text{setify thin}(Q)$  after  $\text{setify}^\circ$ , at  $\text{setify}^\circ \text{ setify} \sqsubseteq 1$  for  $\text{setify}$  simple — dom-laws — then  $\cdot \text{setify} \cdot \text{setify}^\circ$  — triple-chains

$= EA \xrightarrow{\text{thin}(Q)} \text{setify} \xrightarrow{\text{ordered}} LA$

$\text{sort}(P) \triangleq \text{setify}^\circ \text{ ordered } P \triangleq \text{thinlist}$

$\text{sort}(P) \triangleq \text{setify}^\circ \text{ ordered } P \triangleq \text{thinlist}$

$$\text{Map}(f) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{MonoAlg}(f, P) \wedge \text{sortF}(fP f^\circ) \text{listf} \sqsubseteq P(f) \text{sort}(P) \wedge \quad (14.3d)$$

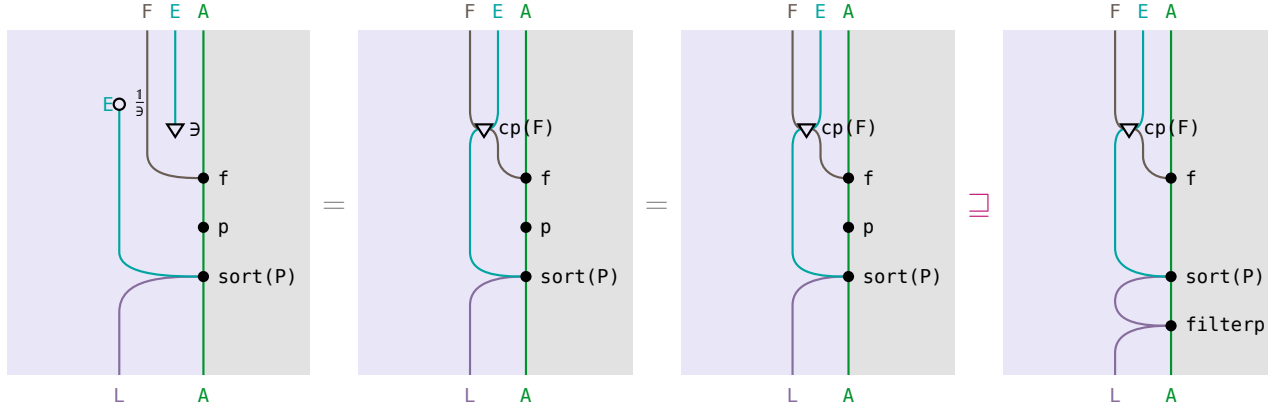
$$\text{sort}(P) \text{filterp} \sqsubseteq E(p) \text{sort}(P) \wedge F(\text{sort}(P)) \text{listcp} \sqsubseteq \text{cp}(F, A) \text{sortF}(F(P)) \Rightarrow F(\text{sort}(P)) \text{listcp} \text{listf}$$

$$\text{filterp} \sqsubseteq \frac{F(\exists)fp}{\exists} \text{sort}(P)$$

one sorted list built from sorted arguments, instead of a set built and then sorted —  $f : FA \rightarrow A$  monotonic on  $P$ ,  $p$  coreflexive,  $F$  linear.

$$\text{Map}(f) \wedge \text{sort}(P) \text{filterp} \sqsubseteq E(p) \text{sort}(P) \Rightarrow \text{cp}(F, A) P(f) \text{sort}(P) \text{filterp} \sqsubseteq \frac{F(\exists)fp}{\exists} \text{sort}(P)$$

sorting the  $f$ -images of the product and filtering by  $p$  is below sorting the set  $F(\exists)fp$  builds



$$\begin{aligned} & \text{FEA} \rightarrow \boxed{\frac{1}{3}} \rightarrow \text{EFEA} \rightarrow \boxed{E(F(\exists)fp)} \rightarrow \text{EA} \rightarrow \boxed{\text{sort}(P)} \rightarrow \text{LA} \\ = & \text{FEA} \rightarrow \boxed{\text{cp}(F, A)} \rightarrow \text{EFA} \rightarrow \boxed{E(fp)} \rightarrow \text{EA} \rightarrow \boxed{\text{sort}(P)} \rightarrow \text{LA} \\ = & \text{FEA} \rightarrow \boxed{\text{cp}(F, A)} \rightarrow \text{EFA} \rightarrow \boxed{P(f)} \rightarrow \boxed{E(p)} \rightarrow \text{EA} \rightarrow \boxed{\text{sort}(P)} \rightarrow \text{LA} \\ \sqsubseteq & \text{FEA} \rightarrow \boxed{\text{cp}(F, A)} \rightarrow \text{EFA} \rightarrow \boxed{P(f)} \rightarrow \text{EA} \rightarrow \boxed{\text{sort}(P)} \rightarrow \boxed{\text{filterp}} \rightarrow \text{LA} \end{aligned}$$

$$\frac{F(\exists)fp}{\exists} = \frac{F(\exists)}{\exists} E(fp) \text{ — pow-laws; } \text{cp}(F) \triangleq \frac{F(\exists)}{\exists} \text{ — thinlist}$$

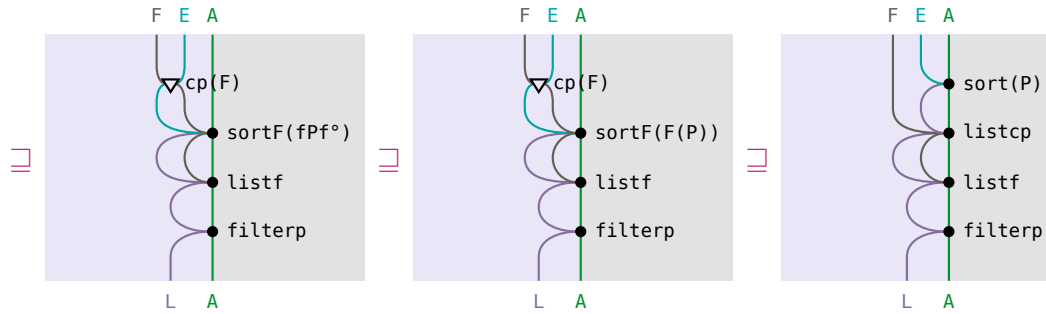
$$E(fp) = E(f)E(p); E(f) = P(f) \text{ for } f \text{ a map — powrel-laws}$$

$$\text{sort}(P) \text{filter}(p) \sqsubseteq E(p) \text{sort}(P) \text{ — (14.3b)}$$

$$\text{Map}(f) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{MonoAlg}(f, P) \wedge \text{sortF}(fP f^\circ) \text{listf} \sqsubseteq P(f) \text{sort}(P) \wedge$$

$$F(\text{sort}(P)) \text{listcp} \sqsubseteq \text{cp}(F, A) \text{sortF}(F(P)) \Rightarrow F(\text{sort}(P)) \text{listcp} \text{listf} \text{filterp} \sqsubseteq \text{cp}(F, A) P(f) \text{sort}(P) \text{filterp}$$

combining the arguments' sorted lists by  $\text{listcp}$ , mapping  $f$  and filtering is below sorting the  $f$ -images and filtering



$$\begin{aligned} \sqsubseteq & \text{FEA} \rightarrow \boxed{\text{cp}(F, A)} \rightarrow \text{EFA} \rightarrow \boxed{\text{sortF}(fP f^\circ)} \rightarrow \text{LFA} \rightarrow \boxed{\text{listf}} \rightarrow \boxed{\text{filterp}} \rightarrow \text{LA} \\ \sqsubseteq & \text{FEA} \rightarrow \boxed{\text{cp}(F, A)} \rightarrow \text{EFA} \rightarrow \boxed{\text{sortF}(F(P))} \rightarrow \text{LFA} \rightarrow \boxed{\text{listf}} \rightarrow \boxed{\text{filterp}} \rightarrow \text{LA} \\ \sqsubseteq & \text{FEA} \rightarrow \boxed{F(\text{sort}(P))} \rightarrow \text{FLA} \rightarrow \boxed{\text{listcp}} \rightarrow \text{LFA} \rightarrow \boxed{\text{listf}} \rightarrow \boxed{\text{filterp}} \rightarrow \text{LA} \end{aligned}$$

$$\text{sort}(fP f^\circ) \text{list}(f) \sqsubseteq P(f) \text{sort}(P) \text{ — (14.3b)}$$

$$FP \sqsubseteq fP f^\circ \text{ — mon-str at } f \text{ a map; } \text{sort}(P) \triangleq \text{setify}^\circ \text{ ordered } P \text{ grows with } P \text{ — thinlist}$$

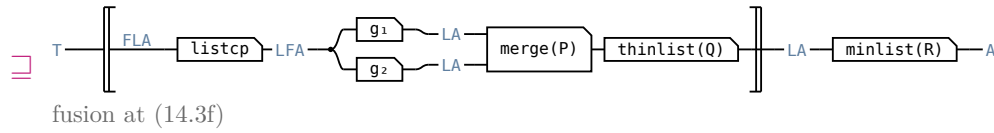
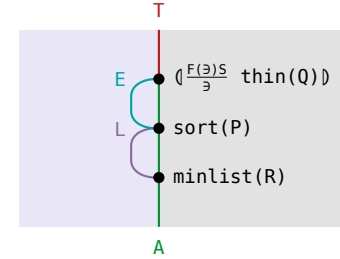
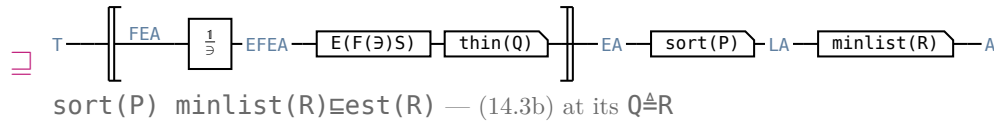
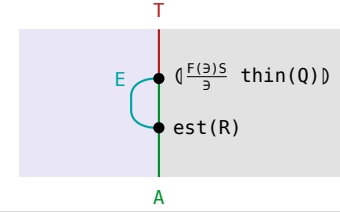
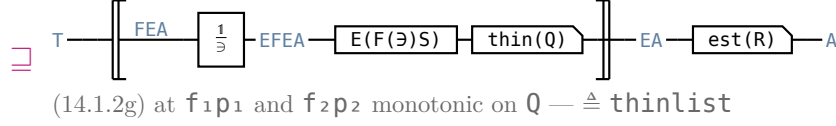
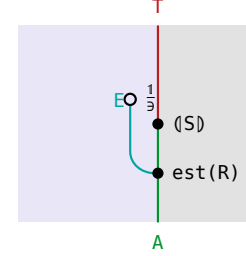
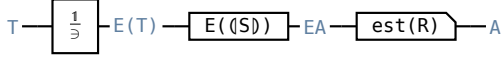
$$F(\text{sort}(P)) \text{listcp} \sqsubseteq \text{cp}(F) \text{sort}(FP) \text{ — (14.3b), } F \text{ linear}$$

$$\begin{aligned}
& \text{PreservesRecip}(F) \wedge \text{Map}(f_1) \wedge \text{Map}(f_2) \wedge Q \sqsubseteq R \wedge 1 \sqsubseteq Q \wedge Q \sqsubseteq Q \wedge R^\circ R^\circ \sqsubseteq R^\circ \wedge \text{MonoAlg}(f_1, p_1, Q) \wedge \\
& \text{MonoAlg}(f_2, p_2, Q) \wedge (\forall X. \forall Y. X \sqsubseteq Y \implies \text{sort}F(X) \sqsubseteq \text{sort}F(Y)) \wedge \text{MonoAlg}(f_1, P) \wedge \text{MonoAlg}(f_2, P) \wedge \\
& \text{sort}F(f_1 \text{ P } f_1^\circ) \text{list}f_1 \sqsubseteq P(f_1) \text{sort}(P) \wedge \text{sort}F(f_2 \text{ P } f_2^\circ) \text{list}f_2 \sqsubseteq P(f_2) \text{sort}(P) \wedge \\
& \text{sort}(P) \text{filter}p_1 \sqsubseteq E(p_1) \text{sort}(P) \wedge \text{sort}(P) \text{filter}p_2 \sqsubseteq E(p_2) \text{sort}(P) \wedge \\
& F(\text{sort}(P)) \text{list}cp \sqsubseteq cp(F, A) \text{sort}F(F(P)) \wedge (\text{sort}(P) \times \text{sort}(P)) \text{merge}(P) \sqsubseteq \text{cup} \text{sort}(P) \wedge \\
& \text{sort}(P) \text{thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{sort}(P) \wedge \text{sort}(P) \text{minlist}(R) \sqsubseteq \text{est}(R) \wedge g_1 = \text{list}f_1 \text{ filter}p_1 \wedge \\
& g_2 = \text{list}f_2 \text{ filter}p_2 \wedge S = f_1 \text{ p}_1 \cup f_2 \text{ p}_2 \implies (\text{list}cp(g_1, g_2) \text{merge}(P) \text{thinlist}(Q)) \text{minlist}(R) \sqsubseteq \frac{(S)}{3} \\
& \text{est}(R)
\end{aligned} \tag{14.3e}$$

a fold on sorted lists of partial solutions, thinned at every step — at  $\triangleq \text{thinlist}$ 's binary thinning data.

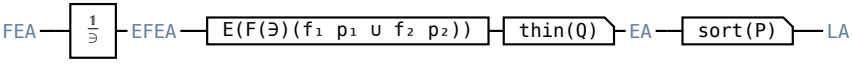
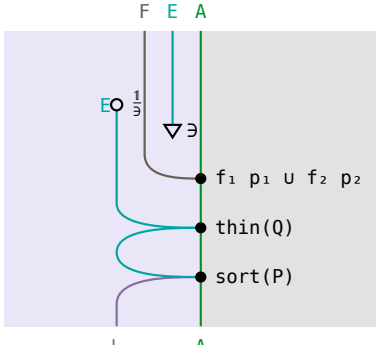
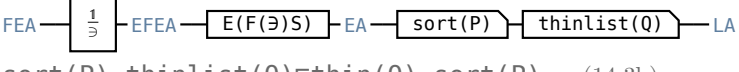
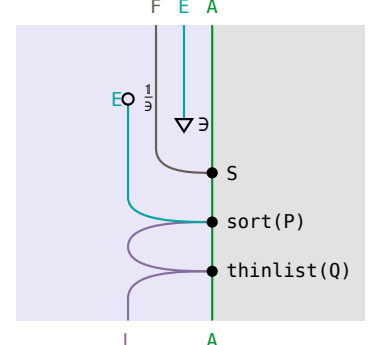
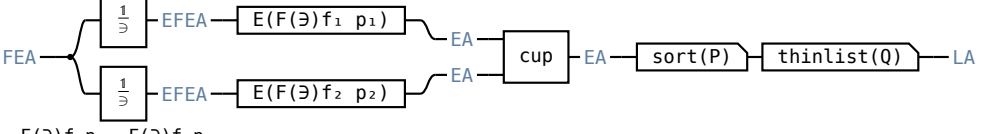
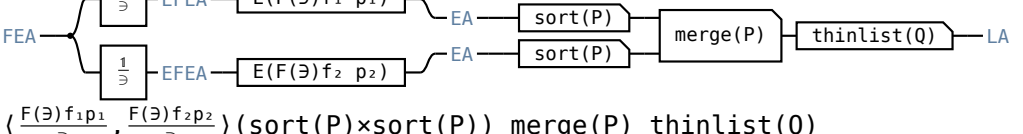
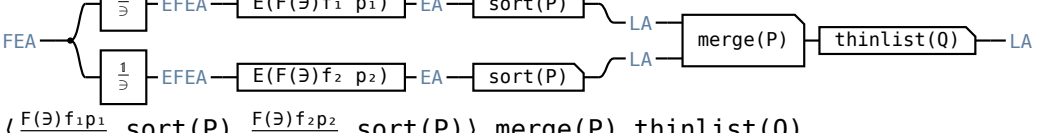
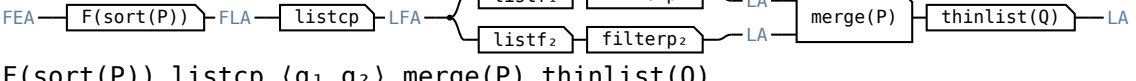
circuit

Hinze–Marsden



$$\begin{aligned}
& \text{Map}(f_1) \wedge \text{Map}(f_2) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{MonoAlg}(f_1, P) \wedge \text{MonoAlg}(f_2, P) \wedge \quad (14.3f) \\
& \text{sortF}(f_1 \text{ Pf}_1^\circ) \text{listf}_1 \sqsubseteq \text{P}(f_1) \text{sort}(P) \wedge \text{sortF}(f_2 \text{ Pf}_2^\circ) \text{listf}_2 \sqsubseteq \text{P}(f_2) \text{sort}(P) \wedge \\
& \text{sort}(P) \text{filterp}_1 \sqsubseteq \text{E}(p_1) \text{sort}(P) \wedge \text{sort}(P) \text{filterp}_2 \sqsubseteq \text{E}(p_2) \text{sort}(P) \wedge \\
& \text{F}(\text{sort}(P)) \text{listcp} \sqsubseteq \text{cp}(\text{F}, \text{A}) \text{sortF}(\text{F}(P)) \wedge (\text{sort}(P) \times \text{sort}(P)) \text{merge}(P) \sqsubseteq \text{cup} \text{sort}(P) \wedge \\
& \text{sort}(P) \text{thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{sort}(P) \Rightarrow \text{F}(\text{sort}(P)) \text{listcp}(\text{listf}_1 \text{ filterp}_1, \text{listf}_2 \text{ filterp}_2) \\
& \text{merge}(P) \text{thinlist}(Q) \sqsubseteq \frac{\text{F}(\exists)(f_1 \text{ p}_1 \cup f_2 \text{ p}_2)}{\exists} \text{thin}(Q) \text{sort}(P)
\end{aligned}$$

sorting the candidate set is what turns the thinning algebra into an algebra on lists — the side condition of binary thinning theorem's last step.

| circuit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Hinze–Marsden                                                                        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| $\sqsubseteq$  <p><math>\text{sort}(P) \text{thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{sort}(P) \text{ — (14.3b)}</math></p>                                                                                                                                                                                                                                                                                  |  |
| $=$  <p><math>\{ \frac{\text{F}(\exists)f_1 p_1}{\exists}, \frac{\text{F}(\exists)f_2 p_2}{\exists} \} \text{cup} \text{sort}(P) \text{thinlist}(Q)</math><br/> <math>S = (f_1 p_1) \cup (f_2 p_2) \text{ — } \triangleq \text{thinlist, then } \triangleq \text{cup}</math></p>                                                                                                                                |                                                                                      |
| $\sqsubseteq$  <p><math>\{ \frac{\text{F}(\exists)f_1 p_1}{\exists}, \frac{\text{F}(\exists)f_2 p_2}{\exists} \} (\text{sort}(P) \times \text{sort}(P)) \text{merge}(P) \text{thinlist}(Q)</math><br/> <math>(\text{sort}(P) \times \text{sort}(P)) \text{merge}(P) \sqsubseteq \text{cup} \text{sort}(P) \text{ — (14.3b)}</math></p>                                                                          |                                                                                      |
| $=$  <p><math>\{ \frac{\text{F}(\exists)f_1 p_1}{\exists} \text{sort}(P), \frac{\text{F}(\exists)f_2 p_2}{\exists} \text{sort}(P) \} \text{merge}(P) \text{thinlist}(Q)</math><br/> <math>\langle X, Y \rangle (\text{sort}(P) \times \text{sort}(P)) = \langle X \text{sort}(P), Y \text{sort}(P) \rangle \text{ — bdm-prod-laws}</math></p>                                                                   |                                                                                      |
|  <p><math>\sqsubseteq \text{F}(\text{sort}(P)) \text{listcp} \langle g_1, g_2 \rangle \text{merge}(P) \text{thinlist}(Q)</math><br/> <math>(14.3d) \text{ at } f_1, p_1 \text{ and at } f_2, p_2, \text{ then } X \langle g_1, g_2 \rangle \sqsubseteq \langle Xg_1, Xg_2 \rangle \text{ — bdm-prod-laws; } g_i \triangleq \text{list}(f_i) \text{filter}(p_i) \text{ — } \triangleq \text{thinlist}</math></p> |                                                                                      |

## §14.4. The knapsack problem

(14.4a)

$\text{vol}, \text{wt} : \text{Item} \rightarrow \text{Real}, \text{value} \triangleq \text{list}(\text{vol}) \text{ sum}, \text{weight} \triangleq \text{list}(\text{wt}) \text{ sum}.$

$\text{subseq} = ([\text{nil}, \text{cons}] \cup [\text{nil}, \pi_2]) \circ \text{w}$ , within  $w$  the coreflexive on  $xs$  with weight  $xs \leq w$ ,  $0 \leq w$ .

$R \triangleq \text{value} \geq \text{value}^\circ$ ,  $Q \triangleq R \cap (\text{weight} \leq \text{weight}^\circ)$ ,  $P \triangleq R \cup Q$ .

$\text{FA} = 1 + \text{Item} \times \text{A}$ ,  $\text{listcp} = \text{wrap} + \text{cpr}$ ,  $g_1 \triangleq \text{list}([\text{nil}, \text{cons}])$   $\text{filter}(\text{within } w) = [\text{list}(\text{nil}), h_1]$ ,  
 $g_2 \triangleq \text{list}([\text{nil}, \pi_2]) = [\text{list}(\text{nil}), h_2].$

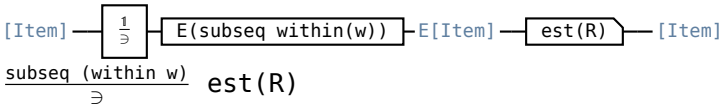
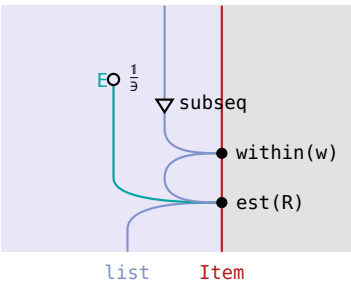
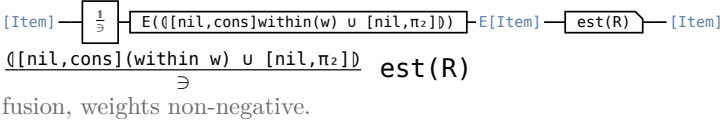
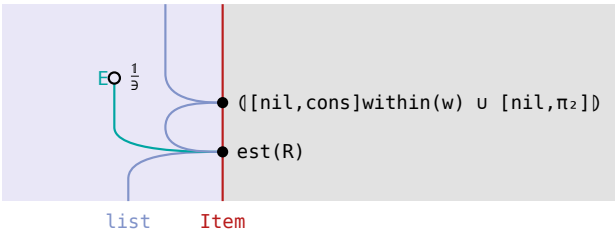
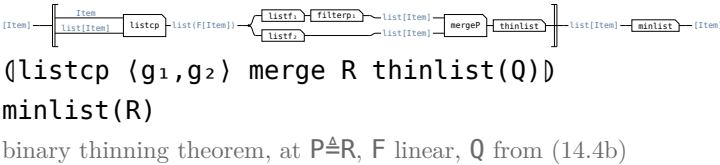
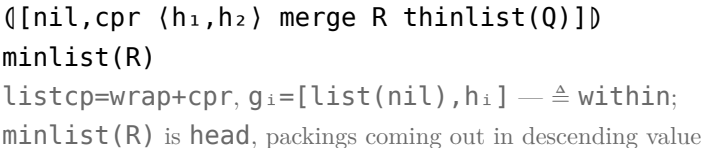
$h_1 \triangleq \text{list}(\text{cons}) \text{filter}(\text{within } w)$ ,  $h_2 \triangleq \text{list}(\pi_2).$

(14.4b)

| the law                                                                                                                                      | what it says                                                                                   |
|----------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| $(1 \times R) (\text{cons } (\text{within } w)) \sqsubseteq \text{cons } (\text{within } w) R$<br>FALSE                                      | not monotonic on R: a selection of greater value need not still fit once one more item goes in |
| $(1 \times Q) (\text{cons } (\text{within } w)) \sqsubseteq \text{cons } (\text{within } w) Q$<br>$(1 \times Q) \pi_2 \sqsubseteq \pi_2 Q$ , | both halves are monotonic on Q once ties in value are broken by weight                         |

$$\begin{aligned}
 & 0 \leq w \wedge (\forall i. 0 \leq \text{wt}(i)) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \\
 & \text{sortF}([\text{nil}, \text{cons}]R[\text{nil}, \text{cons}]^\circ) \text{listf}_1 \sqsubseteq P([\text{nil}, \text{cons}]) \text{sortP} \wedge \\
 & \text{sortF}([\text{nil}, \pi_2]R[\text{nil}, \pi_2]^\circ) \text{listf}_2 \sqsubseteq P([\text{nil}, \pi_2]) \text{sortP} \wedge \text{sortP } \text{filterp}_1 \sqsubseteq E(\text{within}(w)) \text{sortP} \wedge \\
 & F(\text{sortP}) \text{listcp} \sqsubseteq \text{cp}(F, [\text{Item}]) \text{sortF}(F(R)) \wedge (\text{sortP} \times \text{sortP}) \text{mergeP} \sqsubseteq \text{cup } \text{sortP} \wedge \text{sortP} \\
 & \text{thinlist} \sqsubseteq \text{thin}(Q(\text{vol}, \text{wt})) \text{sortP} \wedge \text{sortP } \text{minlist} \sqsubseteq \text{est}(R) \Rightarrow (\text{listcp}(\text{listf}_1 \text{ filterp}_1, \text{listf}_2) \\
 & \text{mergeP } \text{thinlist}) \text{minlist} \sqsubseteq \frac{\text{subseq } \text{within}(w)}{\supset} \text{est}(R)
 \end{aligned}
 \tag{14.4c}$$

the knapsack problem, as a fold that thins the packings kept at each item

| circuit                                                                             | Hinze–Marsden                                                                         |
|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|    |   |
|  |  |
|  |   |
|  |                                                                                       |

## §14.5. The paragraph problem

(14.5a)

```

Line=list+ Word, Para=list+ Line, FA=Word+Word×A, listcp=wrap+cpr.
new(a,xs)=[[a]]#xs, glue(a,xs)=[[a]#head(xs)]#tail(xs), partition≐([wrap wrap,new u glue]) : list+
Word→Para.

width≐([length,(length×1) plus succ]), 0≤length a, fits w the coreflexive on a line x with width x≤w, ok w the
coreflexive on [x]#xs with width x≤w.

white w x=w-width x, collect≐list(sqr) sum, waste w≐init list(white w) collect.

R≐(waste w)≤(waste w)°, Q≐Rn(head head°), P≐T, .

g1≐list([wrap wrap,new]), g2≐list([wrap wrap,glue]) filter(ok w), start≐wrap wrap wrap, h1≐list(new),
h2≐list(glue) filter(ok w).

```

(14.5b)

| the law                                                                                      | what it says                                                                                                             |
|----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| (1×R) glue⊆glue R FALSE                                                                      | glue is not monotonic on R: the waste of a paragraph depends on its whole first line, so no greedy algorithm solves this |
| (1×Q) new⊆new Q<br>(1×Q) (glue (ok w))⊆glue (ok w)Q<br>cons monotonic on collect≤collect°. , | both halves are monotonic on Q once ties in waste are broken by the first line                                           |
| merge T=cat; P≐head prefix head° also serves                                                 | T needs no sorting at all, and prefix is a linear order on first lines of paragraphs of one input                        |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                      |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| $  \begin{aligned}  & (\forall a. 0 \leq \text{len}(a)) \wedge (\forall a. \text{len}(a) \leq w) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{sortF}([\text{wrap} \\  & \text{wrap}, \text{new}] \tau [\text{wrap} \text{ wrap}, \text{new}]^\circ) \text{listf}_1 \sqsubseteq \text{P}([\text{wrap} \text{ wrap}, \text{new}]) \text{sortP} \wedge \text{sortF}([\text{wrap} \text{ wrap}, \text{glue}] \tau [\text{wrap} \\  & \text{wrap}, \text{glue}]^\circ) \text{listf}_2 \sqsubseteq \text{P}([\text{wrap} \text{ wrap}, \text{glue}]) \text{sortP} \wedge \text{sortP} \text{filterp}_2 \sqsubseteq \text{E}(\text{ok}(w)) \text{sortP} \wedge \\  & \text{F}(\text{sortP}) \text{listcp} \sqsubseteq \text{cp}(\text{F}, \text{list}^+((\text{list}^+ \text{ Word}))) \text{sortF}(\text{F}(\tau)) \wedge (\text{sortP} \times \text{sortP}) \text{mergeP} \sqsubseteq \text{cp} \text{sortP} \wedge \text{sortP} \\  & \text{thinlist} \sqsubseteq \text{thin}(\text{Q}(\text{len}, w)) \text{sortP} \wedge \text{sortP} \text{minlist} \sqsubseteq \text{est}(\text{R}) \Rightarrow (\text{listcp}(\text{listf}_1, \text{listf}_2 \text{filterp}_2) \\  & \text{mergeP} \text{thinlist}) \text{minlist} \sqsubseteq \frac{\text{partition fits}(w)}{\supset} \text{est}(\text{R})  \end{aligned}  $ <p>a paragraph laid out as a fold that thins the layouts kept at each word</p> |                      |
| <b>circuit</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Hinze–Marsden</b> |
| $  \begin{aligned}  & \text{list}^+(\text{Word}) \xrightarrow{\frac{1}{3}} \text{E}(\text{list}^+(\text{Word})) \xrightarrow{\text{E}(\text{partition fits}(w))} \text{E}(\text{list}^+(\text{list}^+(\text{Word}))) \xrightarrow{\text{est}(\text{R})} \text{list}^+(\text{list}^+(\text{Word})) \\  & \frac{\text{partition list}^+(\text{fits } w)}{\supset} \text{est}(\text{R})  \end{aligned}  $                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                      |
| $  \begin{aligned}  & \text{list}^+(\text{Word}) \xrightarrow{\frac{1}{3}} \text{E}(\text{list}^+(\text{Word})) \xrightarrow{\text{E}([\text{wrap wrap}, \text{new}] \cup [\text{wrap wrap}, \text{glue}] \text{ok}(w))} \text{E}(\text{list}^+(\text{list}^+(\text{Word}))) \xrightarrow{\text{est}(\text{R})} \text{list}^+(\text{list}^+(\text{Word})) \\  & = \frac{([\text{wrap wrap}, \text{new}] \cup [\text{wrap wrap}, \text{glue}] \text{ok}(w))}{\supset} \text{est}(\text{R})  \end{aligned}  $ <p>fusion, every word fits on a line by itself.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                      |
| $  \begin{aligned}  & \text{list}^+(\text{Word}) \xrightarrow{\frac{1}{3}} \text{E}(\text{list}^+(\text{Word})) \xrightarrow{\text{E}([\text{wrap wrap}, \text{new}] \cup [\text{wrap wrap}, \text{glue}] \text{ok}(w))} \text{E}(\text{list}^+(\text{list}^+(\text{Word}))) \xrightarrow{\text{est}(\text{R})} \text{list}^+(\text{list}^+(\text{Word})) \\  & = \frac{([\text{wrap wrap}, \text{new}] \cup ([\text{wrap wrap}, \text{glue}] \text{ok } w))}{\supset} \text{est}(\text{R})  \end{aligned}  $ <p>the algebra as <math>(f_1 p_1) \cup (f_2 p_2)</math>, <math>p_1 \triangleq 1 \text{ — } \triangleq</math> thinlist.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                      |
| $  \begin{aligned}  & \text{listcp}(\text{g}_1, \text{g}_2) \text{cat thinlist}(\text{Q}) \\  & \sqsubseteq \text{minlist}(\text{R})  \end{aligned}  $ <p>binary thinning theorem, at <math>\text{P} \triangleq \text{T}</math> with merge <math>\tau = \text{cat}</math>, <math>\text{Q}</math> from (14.5b)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                      |
| $  \begin{aligned}  & ([\text{start}, \text{cpr}(\text{h}_1, \text{h}_2) \text{cat thinlist}(\text{Q})]) \\  & = \text{minlist}(\text{R}) \\  & \text{listcp} = \text{wrap} + \text{cpr}, \text{g}_i \text{ along the coproduct — } \triangleq \\  & \text{partition}  \end{aligned}  $                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                      |

## §14.6. Bitonic tours

(14.6a)

$\text{FA} = (\text{City} \times \text{City}) + (\text{City} \times \text{A})$ , the base functor of cons-lists of length at least two;  $\text{listcp} = \text{wrap} + \text{cpr}$ ;  $\text{tc} : \text{City} \times \text{City} \rightarrow \text{Real}$ , neither positive nor symmetric.

$\text{start } (a, b) = ([a, b], [a, b])$ ,  $\text{dropl } (a, ([b] \# \text{xs}, \text{ys})) = ([a] \# \text{xs}, [a] \# \text{ys})$ ,  $\text{dropr } (a, (\text{xs}, [b] \# \text{ys})) = ([a] \# \text{xs}, [a] \# \text{ys})$ ,  $\text{tour} \triangleq ([\text{start}, \text{dropl} \cup \text{dropr}])$ .

$\text{cost } (\text{xs}, \text{ys}) = \text{outcost } \text{xs} + \text{incost } \text{ys}$ ,  $\text{outcost } [a_0, \dots, a_n] = \text{tc } (a_0, a_1) + \dots + \text{tc } (a_{n-1}, a_n)$ ,  $\text{incost } [a_0, \dots, a_n] = \text{tc } (a_1, a_0) + \dots + \text{tc } (a_n, a_{n-1})$ .

$\text{next} \triangleq \text{tail head}$ ,  $\text{next2} \triangleq \text{next} \times \text{next}$ ,  $\text{head2} \triangleq \text{head} \times \text{head}$ ,  $\text{R} \triangleq \text{cost} \leq \text{cost}^\circ$ ,  $\text{Q} \triangleq \text{Rn}(\text{next2 next2}^\circ) \text{n}(\text{head2 head2}^\circ)$ ,  $\text{P} \triangleq \text{T}$ ,  $\text{g}_1 \triangleq \text{list}([\text{start}, \text{dropl}])$ ,  $\text{g}_2 \triangleq \text{list}([\text{start}, \text{dropr}])$ .

(14.6b)

| the law                                                                                                                                                      | what it says                                                                                                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| $(1 \times \text{R}) \text{ dropl} \sqsubseteq \text{dropl } \text{R}$ FALSE<br>$(1 \times \text{R}) \text{ dropr} \sqsubseteq \text{dropr } \text{R}$ FALSE | neither drop is monotonic on R: the two edges it adds and removes depend on head and next of both lists            |
| $(1 \times \text{Q}) \text{ dropl} \sqsubseteq \text{dropl } \text{Q}$<br>$(1 \times \text{Q}) \text{ dropr} \sqsubseteq \text{dropr } \text{Q}$             | both are, once ties in cost are broken by the two second cities — the heads already agree among tours of one input |

| $ \begin{aligned} & (\forall X. \forall Y. X \sqsubseteq Y \implies \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \\ & \text{sortF}([\text{start}, \text{droplFn}] \sqcup [\text{start}, \text{droplFn}]^\circ) g_1 \sqsubseteq P([\text{start}, \text{droplFn}]) \text{sortP} \wedge \\ & \text{sortF}([\text{start}, \text{droprFn}] \sqcup [\text{start}, \text{droprFn}]^\circ) g_2 \sqsubseteq P([\text{start}, \text{droprFn}]) \text{sortP} \wedge \\ & F(\text{sortP}) \text{listcp} \sqsubseteq \text{cp}(F, \text{dTour}(\text{City})) \text{sortF}(F(\tau)) \wedge (\text{sortP} \times \text{sortP}) \text{cat} \sqsubseteq \text{cup} \text{sortP} \wedge \text{sortP} \\ & \text{thinlist} \sqsubseteq \text{thin}(Q(\text{tc})) \text{sortP} \wedge \text{sortP} \text{minlist} \sqsubseteq \text{est}(R) \implies (\text{listcp}(g_1, g_2) \text{cat} \text{thinlist}) \text{minlist} \sqsubseteq \\ & \quad \underbrace{(\text{tourAlg})}_{\exists} \text{est}(R) \\ & \text{a least-cost bitonic tour, as a fold that thins the tours kept at each city} \end{aligned} $ <span style="float: right;">(14.6c)</span> |               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| circuit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Hinze–Marsden |
| $ \begin{aligned} & \text{dCL}(\text{City} \times \text{City}, \text{City}) \xrightarrow{\frac{1}{\exists}} E(\text{dCL}(\text{City} \times \text{City}, \text{City})) \xrightarrow{E(\text{tourAlg})} E(\text{Journey}(\text{City})^2) \xrightarrow{\text{est}(R)} \text{Journey}(\text{City}) \\ & \frac{\text{tour}}{\exists} \text{est}(R) \end{aligned} $                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |               |
| $ \begin{aligned} & \text{dCL}(\text{City} \times \text{City}, \text{City}) \xrightarrow{\frac{1}{\exists}} E(\text{dCL}(\text{City} \times \text{City}, \text{City})) \xrightarrow{E([\text{start}, \text{droplFn}] \sqcup [\text{start}, \text{droprFn}])} E(\text{Journey}(\text{City})^2) \xrightarrow{\text{est}(R)} \text{Journey}(\text{City}) \\ & = \frac{([\text{start}, \text{dropl} \sqcup \text{dropr}])}{\exists} \text{est}(R) \\ & \text{tour} \triangleq ([\text{start}, \text{dropl} \sqcup \text{dropr}]) \triangleq \text{tour} \end{aligned} $                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |               |
| $ \begin{aligned} & \text{dCL}(\text{City} \times \text{City}, \text{City}) \xrightarrow{\frac{1}{\exists}} E(\text{dCL}(\text{City} \times \text{City}, \text{City})) \xrightarrow{E([\text{start}, \text{droplFn}] \sqcup [\text{start}, \text{droprFn}])} E(\text{Journey}(\text{City})^2) \xrightarrow{\text{est}(R)} \text{Journey}(\text{City}) \\ & \sqsubseteq (\text{listcp}(g_1, g_2) \text{cat} \text{thinlist}(Q)) \text{minlist}(R) \\ & \text{binary thinning theorem, at } P \triangleq T \text{ with merge } T = \text{cat}, Q \text{ from (14.6b)} \end{aligned} $                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |               |
| $ \begin{aligned} & ([\text{start wrap}, \text{cpr}(\text{list}(\text{dropl}), \text{list}(\text{dropr})) \text{cat} \\ & = \text{thinlist}(Q)] \text{minlist}(R) \\ & \text{listcp} = \text{wrap} + \text{cpr}, g_i = [\text{list}(\text{start}), \text{list}(\text{drop}_i)] \triangleq \text{tour}; \\ & \text{quadratic, two tours added per step} \end{aligned} $                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |               |